15288-FUNDAMENTOS DE PROGRAMACIÓN

Actividad:

Bibliotecas en lenguaje C (librerías)

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FUENTES:

https://programacionpro.com/bibliotecas-en-programacion-c-todo-lo-que-necesitas-saber/
https://www.w3schools.com/c/c_ref_string.php
https://es.scribd.com/doc/276784201/Stdlib-h
https://www.tutorialspoint.com/c_standard_library/stdlib_h.htm

¿Qué es una biblioteca en C?

Una biblioteca es un conjunto de funcionalidades predefinidas que pueden ser utilizadas por los programadores para completar tareas de manera mas eficiente. En las bibliotecas el codigo ya esta escrito previamente y puede ser reutilizado en otros códigos.

Librerías más utilizadas.

1. stdio.h

La biblioteca stdio.h es una de las más fundamentales en C, ya que proporciona funciones para entrada y salida estándar. Esto significa que puedes utilizar esta biblioteca para leer datos del teclado, mostrar información en la pantalla y mucho más.

Las funciones que permiten esta biblioteca son:

Function	**Description**
fclose()	Closes a file
feof()	Returns a true value when the position indicator has reached the end of the file
ferror()	Returns a true value if a recent file operation had an error
fgetc()	Returns the ASCII value of a character in a file and advances the position indicator
fgets()	Reads a line from a file and advances the position indicator
fopen()	Opens a file and returns a file pointer for use in file handling functions
fprintf()	Writes a formatted string into a file
fputc()	Writes a character into a file and advances the position indicator
fputs()	Writes a string into a file and advances the position indicator
fread()	Reads data from a file and writes it into a block of memory
fscanf()	Reads formatted data from a file and writes it into a number of memory locations
fseek()	Moves the position indicator of a file pointer
ftell()	Returns the value of the position indicator of a file pointer
fwrite()	Writes data from a block of memory into a file
getc()	The same as fgetc()
getchar()	Reads one character of user input and returns its ASCII value
printf()	Writes a formatted string to the console
putc()	The same as fputc()
putchar()	Outputs a single character to the console
puts()	Outputs a string to the console
remove()	Deletes a file
rename()	Changes the name of a file
rewind()	Moves the position indicator to the beginning of the file
scanf()	Reads formatted data from user input and writes it into a number of memory locations
snprintf()	Writes a formatted string into a `char` array (memory-safe)
sprintf()	Writes a formatted string into a `char` array
sscanf()	Reads a formatted string from a `char` array and writes it into a number of memory locations

2. math.h

Si necesitas realizar operaciones matemáticas en tus programas en C, la biblioteca math.h es tu mejor aliada. Contiene funciones para cálculos trigonométricos, exponenciales, logarítmicos y más.

Las funciones que permite son:

Function	Description
acos(x)	Returns the arccosine of x, in radians
acosh(x)	Returns the hyperbolic arccosine of x
asin(x)	Returns the arcsine of x, in radians
asinh(x)	Returns the hyperbolic arcsine of x
atan(x)	Returns the arctangent of x as a numeric value between -PI/2 and PI/2 radians
atan2(y, x)	Returns the angle theta from the conversion of rectangular coordinates (x, y) to polar coordinates (r, theta)
atanh(x)	Returns the hyperbolic arctangent of x
cbrt(x)	Returns the cube root of x
ceil(x)	Returns the value of x rounded up to its nearest integer
copysign(x, y)	Returns the first floating point x with the sign of the second floating point y
cos(x)	Returns the cosine of x (x is in radians)
cosh(x)	Returns the hyperbolic cosine of x
exp(x)	Returns the value of E^x
exp2(x)	Returns the value of 2^x
expm1(x)	Returns e^x-1
erf(x)	Returns the value of the error function at x
erfc(x)	Returns the value of the complementary error function at x
fabs(x)	Returns the absolute value of x
fdim(x)	Returns the positive difference between x and y
floor(x)	Returns the value of x rounded down to its nearest integer
fma(x, y, z)	Returns x*y+z without losing precision
fmax(x, y)	Returns the highest value of a floating x and y
fmin(x, y)	Returns the lowest value of a floating x and y
fmod(x, y)	Returns the floating point remainder of x/y

frexp(x, y)	With x expressed as $m*2^n$, returns the value of *m* (a value between 0.5 and 1.0) and writes the value of *n* to the memory at the pointer y
hypot(x, y)	Returns sqrt(x^2 +y^2) without intermediate overflow or underflow
ilogb(x)	Returns the integer part of the floating-point base logarithm of x
ldexp(x, y)	Returns $x*2^y$
lgamma(x)	Returns the logarithm of the absolute value of the gamma function at x
llrint(x)	Rounds x to a nearby integer and returns the result as a long long integer
llround(x)	Rounds x to the nearest integer and returns the result as a long long integer
log(x)	Returns the natural logarithm of x
log10(x)	Returns the base 10 logarithm of x
log1p(x)	Returns the natural logarithm of x+1
log2(x)	Returns the base 2 logarithm of the absolute value of x
logb(x)	Returns the floating-point base logarithm of the absolute value of x
lrint(x)	Rounds x to a nearby integer and returns the result as a long integer
lround(x)	Rounds x to the nearest integer and returns the result as a long integer
modf(x, y)	Returns the decimal part of x and writes the integer part to the memory at the pointer y
nan(s)	Returns a NaN (Not a Number) value
nearbyint(x)	Returns x rounded to a nearby integer
nextafter(x, y)	Returns the closest floating point number to x in the direction of y
nexttoward(x, y)	Returns the closest floating point number to x in the direction of y
pow(x, y)	Returns the value of x to the power of y
remainder(x, y)	Return the remainder of x/y rounded to the nearest integer
remquo(x, y, z)	Calculates x/y rounded to the nearest integer, writes the result to the memory at the pointer z and returns the remainder.
rint(x)	Returns x rounded to a nearby integer
round(x)	Returns x rounded to the nearest integer
scalbln(x, y)	Returns $x*R^y$ (R is usually 2)
scalbn(x, y)	Returns $x*R^y$ (R is usually 2)
sin(x)	Returns the sine of x (x is in radians)
sinh(x)	Returns the hyperbolic sine of x
sqrt(x)	Returns the square root of x
tan(x)	Returns the tangent of x (x is in radians)
tanh(x)	Returns the hyperbolic tangent of x
tgamma(x)	Returns the value of the gamma function at x
trunc(x)	Returns the integer part of x

3. stdlib.h

stdlib.h es el archivo de cabecera de la biblioteca estándar de C que contiene prototipos de funciones para gestión de memoria dinámica, control de procesos y otras funciones. Incluye funciones para conversión, memoria, control de procesos, ordenación, búsqueda y matemáticas. También define constantes como NULL, y tipos de datos como size_t y structs div_t y ldiv_t.

Las funciones que permiten son:

Function	Description
abs()	Return the absolute (positive) value of a whole number
atof()	Return a `double` value from a string representation of a number
atoi()	Return an `int` value from a string representation of a whole number
atol()	Return a `long int` value from a string representation of a whole number
atoll()	Return a `long long int` value from a string representation of a whole number
calloc()	Allocate dynamic memory and fill it with zeroes
div()	Return the quotient and remainder of an integer division
exit()	End the program
free()	Deallocate dynamic memory
malloc()	Allocate dynamic memory
qsort()	Sort the contents of an array
rand()	Generate a random integer
realloc()	Reallocate dynamic memory
srand()	Initialize the random number generator

4. string.h

La biblioteca string.h es esencial para trabajar con cadenas de texto en C. Con funciones para comparar cadenas, copiarlas, concatenarlas y más, esta biblioteca simplifica la manipulación de texto en tus programas.

Las funciones que permite son:

Function	Description
memchr()	Returns a pointer to the first occurrence of a value in a block of memory
memcmp()	Compares two blocks of memory to determine which one represents a larger numeric value
memcpy()	Copies data from one block of memory to another
memmove()	Copies data from one block of memory to another accounting for the possibility that the blocks of memory overlap
memset()	Sets all of the bytes in a block of memory to the same value
strcat()	Appends one string to the end of another
strchr()	Returns a pointer to the first occurrence of a character in a string
strcmp()	Compares the ASCII values of characters in two strings to determine which string has a higher value
strcoll()	Compares the locale-based values of characters in two strings to determine which string has a higher value
strcpy()	Copies the characters of a string into the memory of another string
strcspn()	Returns the length of a string up to the first occurrence of one of the specified characters
strerror()	Returns a string describing the meaning of an error code
strlen()	Return the length of a string
strncat()	Appends a number of characters from a string to the end of another string
strncmp()	Compares the ASCII values of a specified number of characters in two strings to determine which string has a higher value
strncpy()	Copies a number of characters from one string into the memory of another string
strpbrk()	Returns a pointer to the first position in a string which contains one of the specified characters
strrchr()	Returns a pointer to the last occurrence of a character in a string
strspn()	Returns the length of a string up to the first character which is not one of the specified characters
strstr()	Returns a pointer to the first occurrence of a string in another string
strtok()	Splits a string into pieces using delimiters
strxfrm()	Convert characters in a string from ASCII encoding to the encoding of the current locale